

FST ASSIGNMENT

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Aim:

To install ASP.NET Core using .NET SDK and run a simple web application program

Installation Steps of ASP.NET Core:

Step 1: Download .NET SDK

Go to the official .NET website:

1. Download the latest .NET SDK (Example: .NET 10.0).
2. Click the downloaded installer file.
3. Click Next → Install → Finish.

Installation will take a few minutes.

Step 2: Verify Installation

1. Open Command Prompt.
2. Type this command to verify: `dotnet --version`

If installed correctly, it will display the version number

Step 3: Create a New ASP.NET Core Project

1. Open **Command Prompt**.
2. Choose a location (example: Desktop)
3. Create a new web application: `dotnet new webapp -n MyFirstAspPro`

Step 4: Navigate to Project Folder

Command: `cd Calcu`

Step 5: Run the Application

Command: `dotnet run`

After building, the output will display:

Now listening on: <https://localhost:5001>

Step 6: View Output in Browser

1. Open any web browser.
2. Type: <https://localhost:5001>

The ASP.NET default webpage will be displayed.

This confirms the application is running successfully.

```
C:\Users\Sipika Dharshini>dotnet --version
10.0.103

C:\Users\Sipika Dharshini>cd desktop

C:\Users\Sipika Dharshini\Desktop>mkdir asp_pro

C:\Users\Sipika Dharshini\Desktop>cd asp_pro

C:\Users\Sipika Dharshini\Desktop\asp_pro>dotnet new webapp -n MyFirstAspPro

Welcome to .NET 10.0!
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SDK Version: 10.0.103

Telemetry
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The .NET tools collect usage data in order to help us improve your experience. It is collected by Microsoft and shared with the community. You can opt-out of
telemetry by setting the DOTNET_CLI_TELEMETRY_OPTOUT environment variable to '1' or 'true' using your favorite shell.

Read more about .NET CLI Tools telemetry: https://aka.ms/dotnet-cli-telemetry

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Installed an ASP.NET Core HTTPS development certificate.
To trust the certificate, run 'dotnet dev-certs https --trust'
Learn about HTTPS: https://aka.ms/dotnet-https

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Write your first app: https://aka.ms/dotnet-hello-world
Find out what's new: https://aka.ms/dotnet-whats-new
Explore documentation: https://aka.ms/dotnet-docs
Report issues and find source on GitHub: https://github.com/dotnet/core
Use 'dotnet --help' to see available commands or visit: https://aka.ms/dotnet-cli
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The template "ASP.NET Core Web App (Razor Pages)" was created successfully.
This template contains technologies from parties other than Microsoft, see https://aka.ms/aspnetcore/10.0-third-party-notices for details.

Processing post-creation actions...
Restoring C:\Users\Sipika Dharshini\Desktop\asp_pro\MyFirstAspPro\MyFirstAspPro.csproj:
Restore succeeded.
```

Sample Code :-

Program.cs:

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.MapGet("/", () => "Welcome to Calculator API - FST Lab");
app.MapGet("/add/{num1}/{num2}", (double num1, double num2) =>
{
    return Results.Ok(new
    {
        Operation = "Addition",
        Number1 = num1,
        Number2 = num2,
        Result = num1 + num2
    });
});
app.MapGet("/subtract/{num1}/{num2}", (double num1, double num2) =>
{
    return Results.Ok(new
    {
        Operation = "Subtraction",
        Number1 = num1,
        Number2 = num2,
        Result = num1 - num2
    });
});
app.MapGet("/multiply/{num1}/{num2}", (double num1, double num2) =>
{
    return Results.Ok(new
    {
```

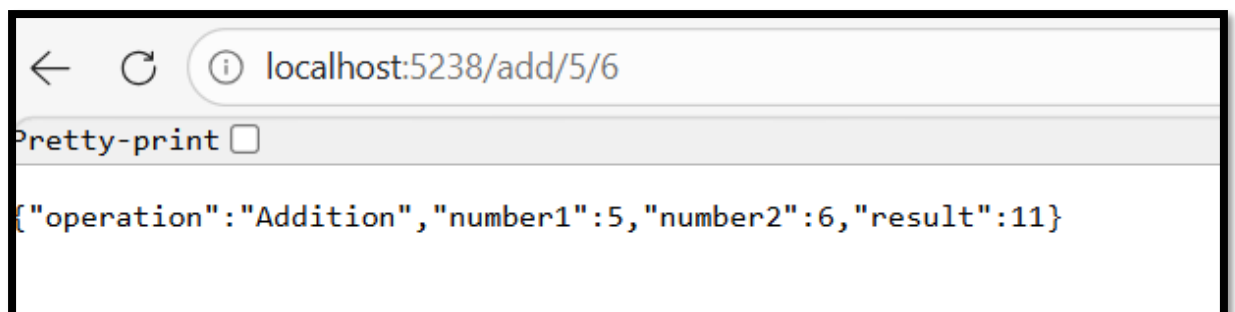
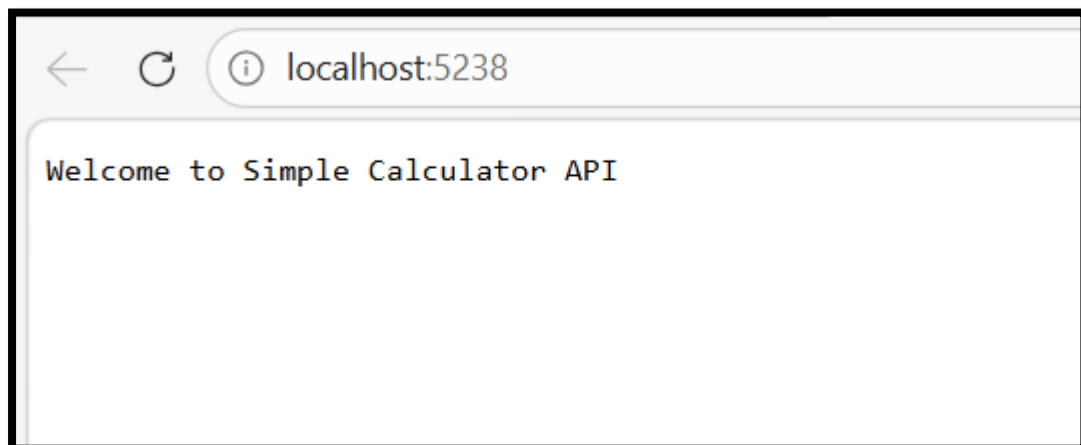
```

        Operation = "Multiplication",
        Number1 = num1,
        Number2 = num2,
        Result = num1 * num2
    });
});
app.MapGet("/divide/{num1}/{num2}", (double num1, double num2) =>
{
    if (num2 == 0)
        return Results.BadRequest("Division by zero is not allowed.");

    return Results.Ok(new
    {
        Operation = "Division",
        Number1 = num1,
        Number2 = num2,
        Result = num1 / num2
    });
});
app.Run();

```

OUTPUT:



```
localhost:5238/subtract/7/6
Pretty-print ☐
{"operation":"Subtraction","number1":7,"number2":6,"result":1}
```

```
localhost:5238/multiply/7/6
Pretty-print ☐
{"operation":"Multiplication","number1":7,"number2":6,"result":42}
```

```
localhost:5238/divide/7/6
Pretty-print ☐
{"operation":"Division","number1":7,"number2":6,"result":1.1666666666666667}
```

```
localhost:5238/divide/7/0
Pretty-print ☐
"Division by zero is not allowed."
```